

Y5 MATHS WEEK-3

PERCENTAGE

Question 1: Work out the following

- | | | |
|--------------------|-----------------------|------------------|
| (a) 10% of 90m | (b) 25% of 24 seconds | (c) 10% of 500kg |
| (d) 50% of 38g | (e) 75% of 60ml | (f) 1% of £400 |
| (g) 25% of 44 days | (h) 50% of 18 days | (i) 75% of 36p |
| (j) 25% of £28 | (k) 1% of \$9100 | (l) 10% of £6 |
| (m) 1% of 80m | (n) 75% of 16 miles | (o) 1% of 70kg |
| (p) 50% of 3.4km | | |

Question 2: Work out the following

- | | | |
|---------------------|------------------|-----------------------|
| (a) 20% of 45km | (b) 5% of £80 | (c) 2% of 900m |
| (d) 30% of 80p | (e) 3% of \$8000 | (f) 40% of 85 seconds |
| (g) 15% of 80 hours | (h) 5% of 24kg | (i) 60% of 40km |
| (j) 30% of £50 | (k) 70% of 800cm | (l) 20% of 15cm |
| (m) 11% of 300m | (n) 26% of 5000m | (o) 55% of £6 |
| (p) 15% of 260kg | | |

Y5 MATHS WEEK-3

Question 3: Work out the following

- | | | |
|---------------------|-----------------------|-------------------|
| (a) 35% of £700 | (b) 6% of 250g | (c) 23% of 400cm |
| (d) 52% of 800m | (e) 61% of 1500 | (f) 7% of 800GB |
| (g) 45% of 400g | (h) 80% of 550 people | (i) 90% of 1350ml |
| (j) 76% of £90,000 | (k) 85% of 80 hours | (l) 12% of £9 |
| (m) 6% of £40 | (n) 11% of 9m | (o) 28% of 4km |
| (p) 71% of 5 tonnes | | |

Question 4: Calculate the following

- | | | |
|-----------------------|-----------------------|------------------|
| (a) 30% of 340m | (b) 5% of 92g | (c) 30% of £360 |
| (d) 35% of 92 seconds | (e) 15% of 700kg | (f) 45% of 8mm |
| (g) 18% of 40 miles | (h) 65% of 200ml | (i) 98% of 9m |
| (j) 55% of 22 points | (k) 20% of 2.5kg | (l) 19% of 700ml |
| (m) 27% of 80g | (n) 63% of 40 seconds | (o) 86% of 6km |
| (p) 92% of 90 litres | | |

PERCENTAGE WORDED PROBLEMS

- Question 1:** A primary school has 320 students.
50% of the students are boys.
How many of the students are boys?
- Question 2:** There are 900 fans at a football match between Rovers and City.
40% of the fans support City.
How many fans support City?
- Question 3:** Hannah is paid £360.
She spends 25% on her rent, 30% on food and bills and saves the rest.
(a) How much does Hannah spend on rent?
(b) How much does Hannah spend on food and bills?
(c) How much does Hannah save?
- Question 4:** There are 240 students in Year 7.
20% cycle to school.
55% are driven to school.
The rest walk to school.
(a) How many students cycle to school?
(b) How many students are driven to school?
(c) How many students walk to school?
- Question 5:** Fredrick is an estate agent in New York and earns 6% commission on every property sold. How much will he receive if he sells a flat for \$750,000?

- Question 6:** A cake weighs 900g.
35% of the cake is sugar.
Work out how many grams of sugar are in the cake.
- Question 7:** There are 500 members of a running club.
42% of these members are male.
Work out 42% of 500.
- Question 8:** Martin gives 45% of £80 to his sister.
How much money does Martin give to his sister?
- Question 9:** Emma is paid £28,000 each year.
She is given a pay rise of 15%.
Work out 15% of £28,000.
- Question 10:** Mrs Jones donates 5% of her salary each year to charity.
She is paid £36,000.
Work out how much money she donates to charity.

- Question 11:** 18% of the people on an island are left handed.
The population of the island is 0.6 million.
Work out how many people are left handed.
- Question 12:** Frank organised a raffle.
He sells 250 tickets for £4 each.
The prizes cost £250.
He gives 60% of the profit to Charity A and 40% of the profit to Charity B.
Work out how much each charity receives.
- Question 13:** Michael is going to buy a car.
The car costs £3000.
He pays a deposit of 25%.
Michael pays the rest of the money over 25 monthly payments.
Work out the cost of each monthly payment.
- Question 14:** An adult ticket for a museum is £12.00
A child ticket costs 60% of the price of an adult ticket.
Mrs Jenkins and her three children go to the museum.
Mrs Jenkins pays with three £20 notes.
How much change will she receive?
- Question 15:** Frances and her family go for a meal while on holiday in Florida.
They are told it is normal to tip 15%.
The meal costs \$150.
Frances tips \$20, is this enough?

PERCENTAGE - MIXED QUESTIONS

1. Calculate 70% of 40
.....
(2)
2. A bag of sugar contains 500g.
A special offer packet contains an extra 20%.
Work out how much extra sugar is in the special offer packet.
.....g
(2)
3. A new car is priced at £8200.
In a sale it is reduced by 15%.
Calculate the reduction in price.
£.....
(2)
4. Calculate 4% of \$700
\$.....
(2)
5. Mary wins £500.
She gives 40% to her mother.
Mary gives 25% to her sister.
She keeps the rest.
Work out how much each person receives.
Mother £.....
Sister £.....
Mary £.....
(3)
6. Barry's salary is £16,000 and he is due to get an increase of 6%.
Calculate how much this increase will be.
£.....
(2)

7. Harry wants to buy a motorbike that costs £720.
He saves 25% of his wages each month.
Each month Harry earns £480.
Calculate how many months it will take Harry to save enough money.
.....
(3)
8. There are 25,000 fans at a football match.
16% of the fans are women.
Work out how many women attended the match.
.....
(2)
9. Calculate 65% of 400
.....
(2)
10. At Frome International train station, 40% of trains were late in a week.
In that week there were 500 trains.
Calculate how many trains were on time.
.....
(3)
11. Work out 96% of 800.
.....
(2)
12. A company is hosting a charity meal.
900 people work for the company.
It is estimated 65% of the employees will attend.
It is expected that if an employee attends, they will bring a partner.
The cost of attending the meal is £6.
How much money will be raised for charity?
£.....
(4)

REVERSE PERCENTAGE

- Question 1:** 25% of all the children in a class are left handed.
6 children are left handed.
How many children are there in the class altogether?
- Question 2:** 35% of the members of a tennis club are pensioners.
42 members are pensioners.
(a) How many members are there in total?
(b) How many members are not pensioners?
- Question 3:** A group of people sit their driving theory test and 30 people passed.
75% of the people passed the driving theory test.
How many people sat the test altogether?
- Question 4:** An energy bar contains 3g of protein.
8% of the bar is protein.
What is the total mass of the bar?
- Question 5:** Bristol is a city in England.
The population of Bristol is 180,000
This population is 6% of the total population of the region.
What is the total population of the region?

- Question 6:** Heather invested money into a savers bank account.
Each year the money in the account earns 10% interest.
After one year, the total amount of money in the account was £2750
How much did Heather invest?
- Question 7:** A chair is on sale at a price of £24.00
This is a 20% reduction of the normal price.
What was the price of the chair before the reduction?
- Question 8:** The population of an island has decreased by 35% over 40 years.
The population in 2018 was 390
What was the population in 1978?
- Question 9:** Sinead buys a watch.
20% VAT is added to the price of the watch.
Sinead then has to pay a total of £72
What is the price of the watch with no VAT added?

ANSWER KEY

Question 1 (Percentage of an amount)

- (a) 9m (b) 6 seconds (c) 50kg (d) 19g
(e) 45ml (f) £4 (g) 11 days (h) 9 days
(i) 27p (j) £7 (k) \$91 (l) £0.60
(m) 0.8m (n) 12 miles (o) 0.7kg (p) 1.7km

Question 2

- (a) 9km (b) £4 (c) 18m (d) 24p
(e) \$240 (f) 34 seconds (g) 12 hours (h) 1.2kg
(i) 24km (j) £15 (k) 560cm (l) 3cm
(m) 33m (n) 1300m (o) £3.30 (p) 39kg

Question 3

- (a) £245 (b) 15g (c) 92cm (d) 416m
(e) 915 (f) 56GB (g) 180g (h) 440 people
(i) 1215ml (j) £68,400 (k) 68 hours (l) £1.08
(m) £2.40 (n) 0.99m (o) 1.12km (p) 3.55 tonnes

Question 4

- (a) 102m (b) 4.6g (c) £108 (d) 32.2 seconds
(e) 105kg (f) 3.6mm (g) 7.2 miles (h) 130ml
(i) 8.82m (j) 12.1 points (k) 0.5kg (l) 133ml
(m) 21.6g (n) 25.2 seconds (o) 5.16km (p) 82.8 litres

ANSWER KEY - Worded Problems

1. 160 boys
2. 360 fans
3. (a) £90 rent (b) £108 food/bills (c) £162 saved
4. (a) 48 cycle (b) 132 driven (c) 60 walk
5. \$45,000
6. 315g sugar
7. 210 male members
8. £36
9. £4,200
10. £1,800
11. 108,000 people
12. Charity A: £450 Charity B: £300
13. £90 per month
14. £26.40 change
15. No - the tip needed is \$22.50, so \$20 is not enough (short by \$2.50)

ANSWER KEY - Mixed Questions & Reverse Percentage

Mixed Questions

1. 28
2. 100g
3. £1,230
4. \$28
5. Mother £200 Sister £125 Mary £175
6. £960
7. 6 months
8. 4,000 women
9. 260
10. 300 trains on time
11. 768
12. £7,020

Reverse Percentage

1. 24 children
2. (a) 120 members total (b) 78 not pensioners
3. 40 people sat the test
4. 37.5g total mass
5. 3,000,000 total population
6. £2,500 invested
7. £30 original price
8. 600 population in 1978
9. £60 price with no VAT